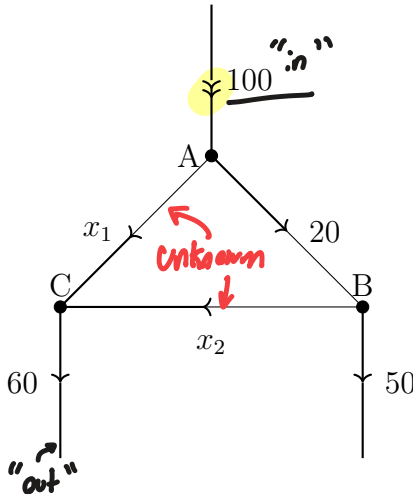


Week 2

Solution to linear system does not always exist:



flow rate conservation
at each node

$$A : 100 = 20 + x_1$$

$$B : 20 = x_2 + 50$$

$$C : x_1 + x_2 = 60$$

↙ solve
for x

$$\text{node A : } 100 = x_1 + 20 \quad \rightarrow \quad x_1 = 80$$

$$\text{node B : } 20 = x_2 + 50 \quad \rightarrow \quad x_2 = -30$$

$$\text{node C : } x_1 + x_2 = 60 \quad \rightarrow \quad 80 + (-30) = +50 \neq 60$$

doesn't work

Reason:

$\therefore$ the system isn't
consistent

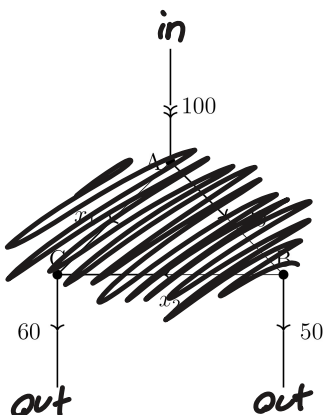
"IN"

100

$\neq$

"OUT"

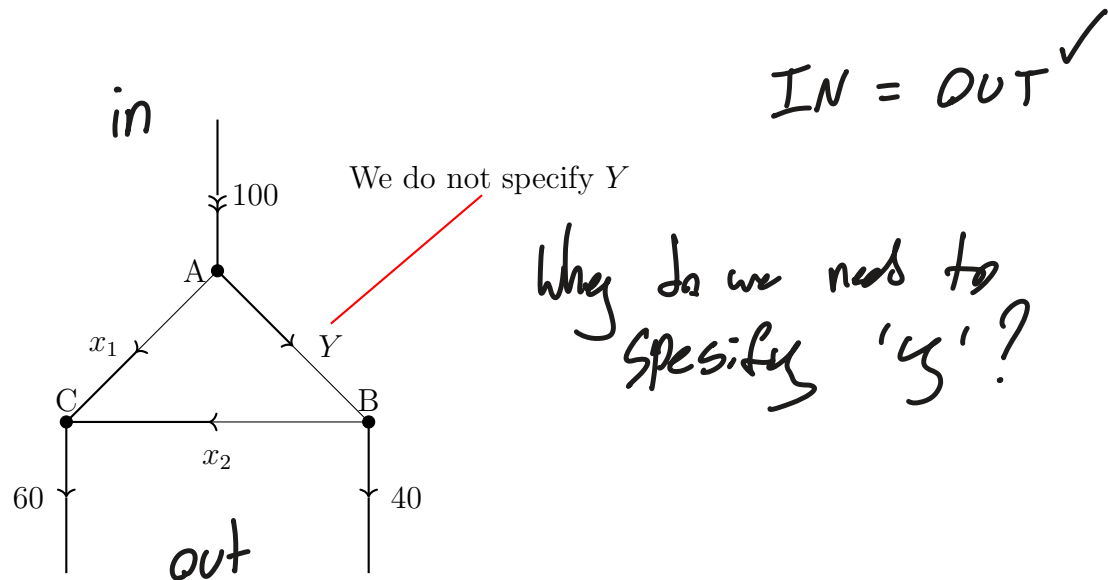
$60 + 50 = 110$



$$100 \neq 50 + 60$$

we can tell it's not
consistent very easy

Solution to linear system is not always unique:



node A : $100 = x_1 + Y \rightarrow x_1 = 100 - Y$

node B : $Y = x_2 + 40 \rightarrow x_2 = Y - 40$

node C : $x_1 + x_2 = (100 - Y) + (Y - 40) = 60 \rightarrow \checkmark \cong 60$

$\Rightarrow$ True for any Y

Consistent?

Pipe size will change
flow rate, $\therefore$ you must state 'Y'

"IN"

"OUT"

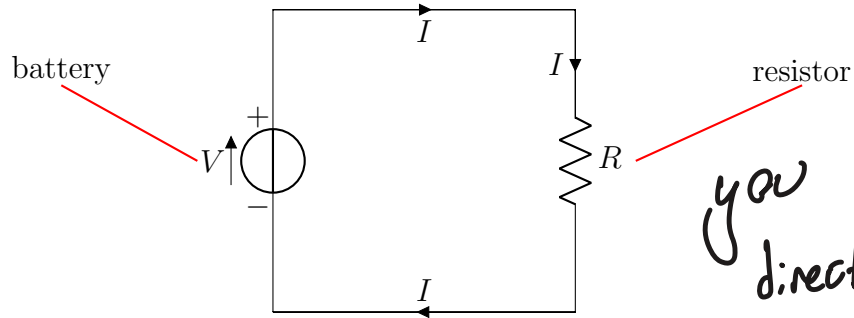
$$100 = 60 + 40 = 100$$

$\Rightarrow$ In the course we study when the system of linear equations is consistent, and when there is a unique solution

$\Rightarrow$ In previous example setting $Y = 20$ gives a unique solution; without specifying Y system of equations is underdetermined

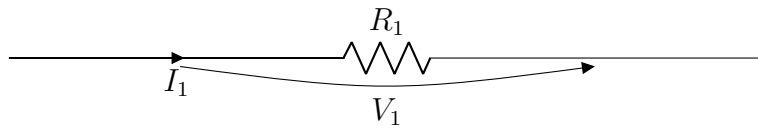
Electric circuits

⇒ Elementary electric circuit: *current*



Amps

- **current** I flows from the positive pole of the battery to the negative pole
- battery is characterized by the **voltage** V (electric potential difference between the poles)
- current is characterized by a flow of positively charged particles (convention) — negatively charged particles (electrons) move in the direction opposite to that of the current
- **resistor** R reduces the *electric potential* according to **Ohm's law**:



$$\begin{array}{ccccc}
 \textcircled{V_1} & = & \textcircled{R_1} & \cdot & I_1 \\
 \text{potential drop } V_1 & & \text{resistance of the resistor } R_1 & & \text{current } I_1 \\
 \text{across the resistor} & & \text{property of the material} & & \text{through the resistor} \\
 \text{in volts (V)} & & \text{in Ohms } (\Omega) & & \text{in amperes (A)}
 \end{array}$$

Along the wire
current cannot change

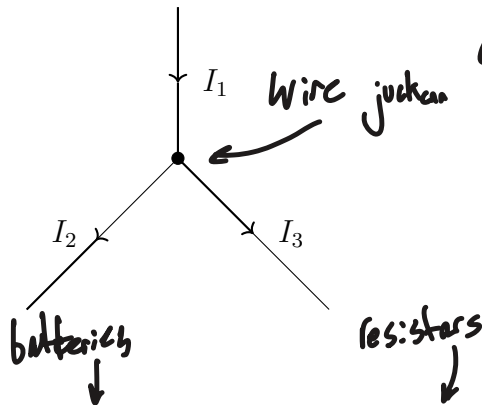
7

review stuff
ez cake

Kirchoff's laws of electric circuits:

① Current conservation at each node

$$I_N = I_{OUT}$$



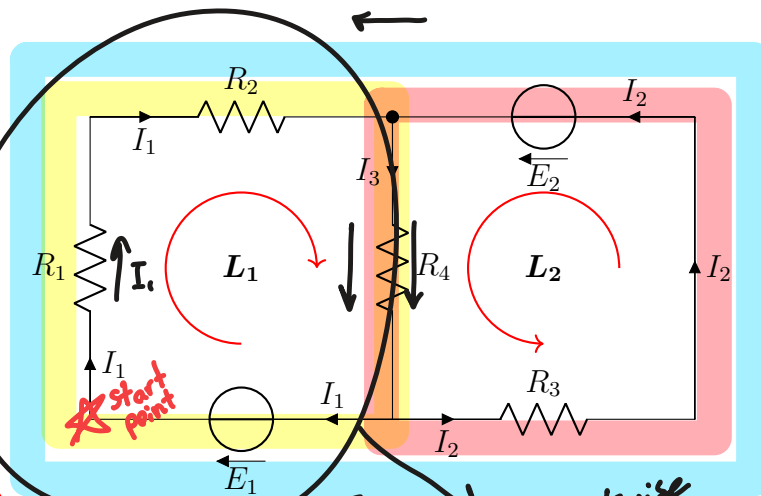
Current can only change at junction

$$\underbrace{I_1}_{\text{"in"}} = \underbrace{I_2 + I_3}_{\text{"out"}}$$

② Sum of voltage rises = sum of voltage drops

⇒ In normal language: electric potential change is zero for any closed loop of the circuit

⇒ you need to choose loop orientation



clockwise

$$\begin{aligned} \text{drops} &= +R_1 I_1 + R_2 I_1 + R_4 I_3 \\ \text{rises} &= +E_1 \end{aligned}$$

$$\hookrightarrow R_1 I_1 + R_2 I_1 + R_4 I_3 = E_1$$

counter clockwise

$$\begin{aligned} \text{drops} &: -I_3 R_4 - I_1 R_2 - I_1 R_1 \\ \text{rises} &: -E_3 \end{aligned}$$

against the battery (-)

$$\hookrightarrow -I_3 R_4 - I_1 R_2 - I_1 R_1 = -E_3$$

⇒ There are 3 closed loops: $\{L_1, L_2, L_3\}$; orientation in red

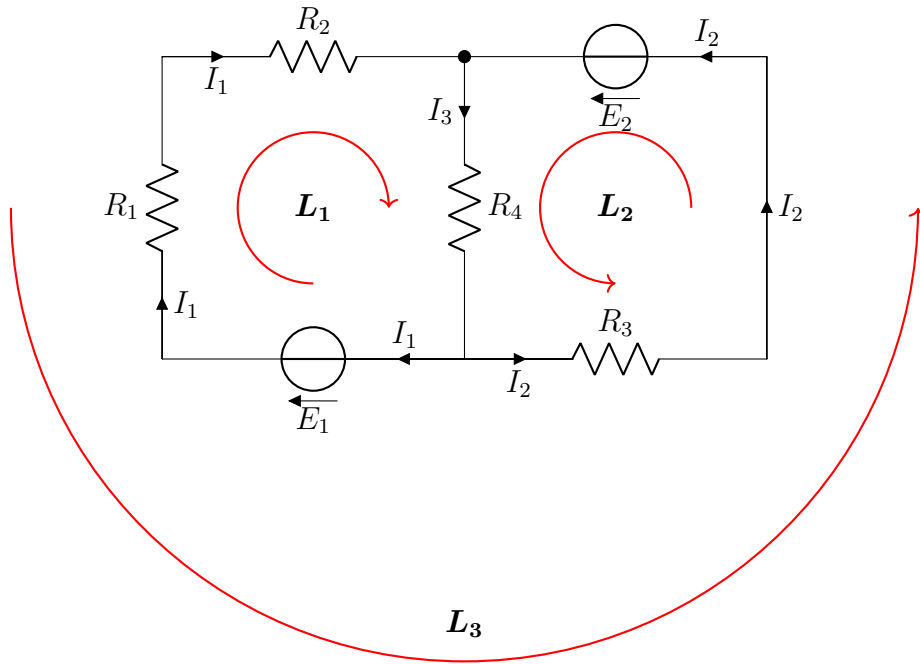
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Same

DNM

Same drops
Same rises

how is this Algebra...



$$\mathbf{L_1 :} \quad \underbrace{R_1 \cdot I_1 + R_2 \cdot I_1 + R_4 \cdot I_3}_{\text{"voltage drops"}} = \underbrace{E_1}_{\text{"voltage rises"}}$$

$$\mathbf{L_2 :} \quad \underbrace{R_4 \cdot I_3 + R_3 \cdot I_2}_{\text{"voltage drops"}} = \underbrace{E_2}_{\text{"voltage rises"}}$$

$$\mathbf{L_3 :} \quad \underbrace{R_3 \cdot I_2 + (-R_2 \cdot I_1) + (-R_1 \cdot I_1)}_{\text{"voltage drops"}} = \underbrace{(-E_1) + E_2}_{\text{"voltage rises"}}$$

$\Rightarrow$ Note **minus** signs: they appear because we are going 'against' the current direction

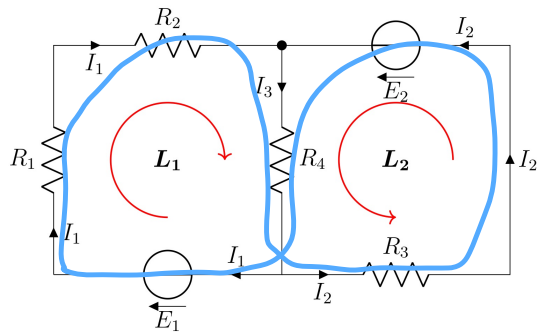
$\Rightarrow$ Note:

$$\mathbf{L_3 = L_2 - L_1}$$

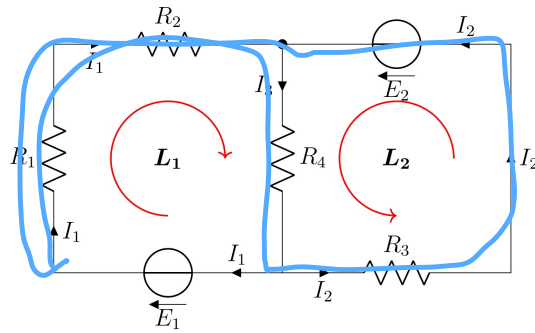
that is, loop $\mathbf{L_3}$ is not *independent*

- orientation } obviously
 - starting point }

$$R_1 I_1 + R_2 I_1 + R_4 I_3 = E_1$$



infinitely many loops

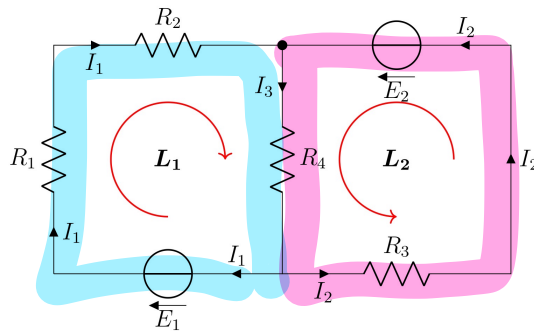


$\infty \rightarrow$ need to pick
"independent loops"

different choice of independent loops
 gives different system of equations

$\Rightarrow$ Solution Same

Loop doesn't
 need a battery

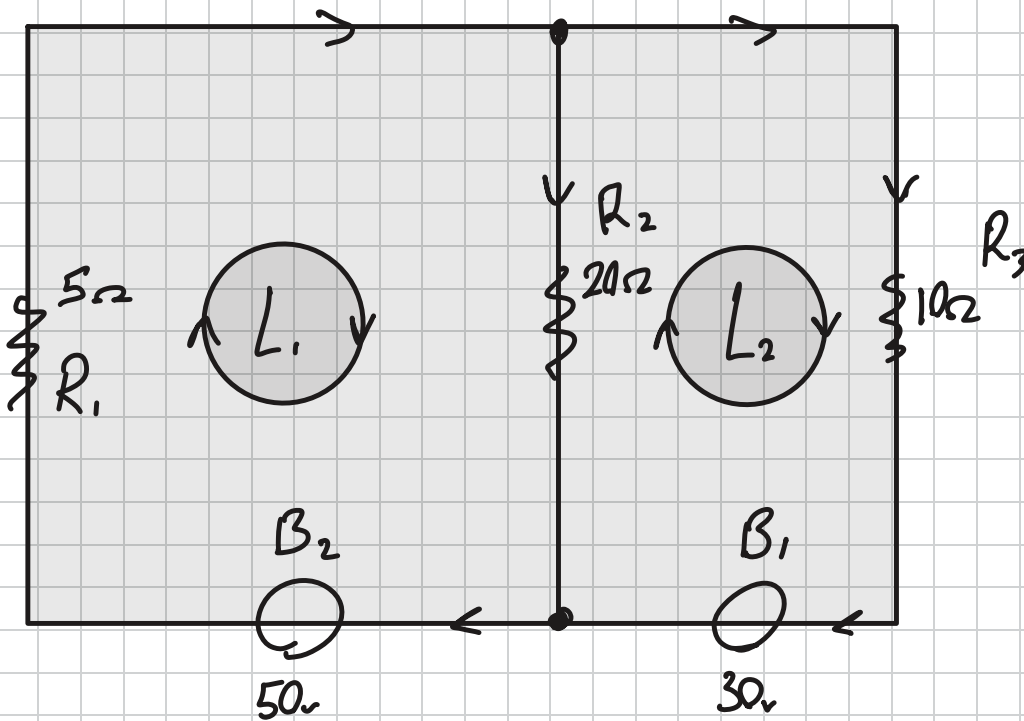


independent loops = circuit parts

$\therefore$ two loops for this example

Very clean 'e'

Can't find
Lesson 3



=> # of currents?

I_1, I_2, I_3 and nothing else

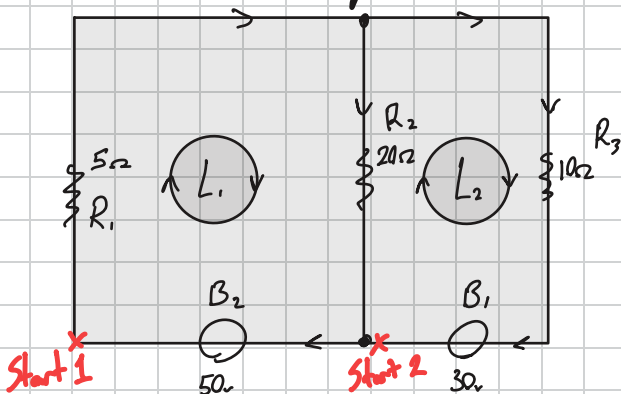
=> direction of current at your disposal

=> Flow diagram $\begin{matrix} \rightarrow \\ \rightarrow \end{matrix} \begin{matrix} 1 \\ \infty \end{matrix}$

E.S. $\rightarrow$ must get 1 solution

=> Pick up equations

of equations = # of currents



of nodes = 2

eq = # of nodes - 1
= 1

$I_1 = I_2 + I_3$

of loop eq = the rest = $3 - 1 = 2$

$$L_2 : -20 I_3 + 10 I_2 = +30$$

$$L_1 : 5 I_1 + 20 I_3 = +50$$

$$\begin{cases} I_1 = I_2 + I_3 \\ -20 I_3 + 10 I_2 = 30 \\ 5 I_1 + 20 I_3 = 50 \end{cases}$$

System of linear equations

